



Thermostat Wiring: Heat Pump

Heat pump field reference | Brackett Heating, Air & Plumbing

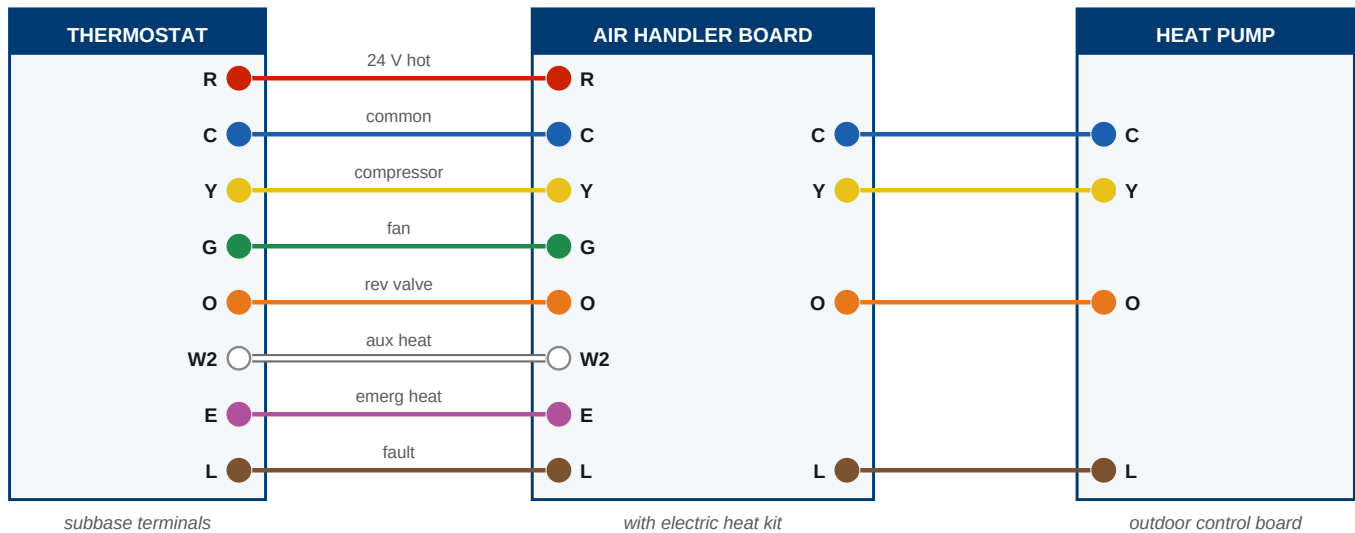
SCOPE

Air-source heat pump with an indoor air handler and electric backup heat. The difference from a conventional system is the reversing valve (O/B) and the auxiliary / emergency heat terminals.

TERMINAL REFERENCE · HEAT PUMP

Term	Common color	What it does	Lands on
R	Red	24 VAC hot from the air handler transformer	Air handler R.
C	Blue	24 VAC common. Powers the thermostat.	Air handler C, and out to the outdoor unit.
Y	Yellow	Compressor contactor.	Air handler Y, passed out to the heat pump.
G	Green	Indoor blower.	Air handler G.
O	Orange	Reversing valve, energized in cooling . This is the majority: Carrier, Trane, Goodman, Lennox, York, most Rheem.	Air handler O, out to the heat pump.
B	Dark blue	Reversing valve, energized in heating . Older Rheem / Ruud and some Amana. Never land both O and B.	Air handler B, out to the heat pump.
W2 / AUX	White	Auxiliary heat - the electric strips. Runs alongside the compressor when the heat pump cannot keep up.	Air handler W2 or AUX.
E	Pink or brown	Emergency heat. Strips only, compressor locked out.	Air handler E. Many air handlers jumper E to W2.
L	Brown	Fault / lockout signal back from the outdoor board to light the stat's service indicator.	Air handler L, out to the heat pump.
Y2	Violet	Second stage compressor on a two-stage heat pump.	Air handler Y2, out to the heat pump.

WIRING DIAGRAM · HEAT PUMP



The thermostat cable lands on the indoor unit board. Only the terminals carried through to the third column run out to the outdoor unit.

O / B IS THE ONE THAT BITES PEOPLE

O energizes the reversing valve in cooling. B energizes it in heating. Land the one the equipment uses, then set the thermostat's O/B configuration to match.

Get it backwards and the system heats when you call for cool and cools when you call for heat. Everything will look correctly wired.

Default to O unless the equipment says otherwise. Verify at the outdoor unit's wiring diagram, not by wire color.

Auxiliary vs. emergency heat

- **Auxiliary (W2 / AUX)** runs *with* the compressor. The thermostat brings it on when the heat pump alone cannot hold setpoint, or during defrost.
- **Emergency (E)** runs *instead of* the compressor. The customer selects it when the heat pump is down. Many air handlers jumper E and W2 to the same sequencer.
- On a heat pump thermostat, do not land the electric heat on plain W. Use W2 or AUX. Wiring strips to W makes the stat treat the system as conventional and the compressor will never run in heat.
- Confirm the stat is configured for the correct number of auxiliary stages, and that the compressor lockout / balance point is set if the customer wants one.

Dual fuel (heat pump over a gas furnace)

- Requires either a dual-fuel capable thermostat or a dual-fuel changeover kit. A standard heat pump stat will not do it safely.
- The compressor and the gas furnace must **never** run at the same time. Gas heat raises the coil temperature and will trip the high pressure switch.
- Set the outdoor balance point so the furnace takes over below the temperature where the heat pump stops being economical.
- An outdoor temperature sensor is normally required. Mount it on the north side, out of direct sun and away from the condenser discharge.

Safety and shop rules

- Kill power at the disconnect and the furnace service switch before landing or removing any wire. The 24 V side is still live if the transformer is powered.
- Shorting R to C blows the control transformer or the board fuse. Most boards carry a 3 A or 5 A automotive blade fuse - carry spares.
- Wire colors are convention, not code. Ring out every conductor before you trust it. On a change-out, label the old wires at both ends before you pull the old stat.
- Land a C wire on every modern thermostat. Power-stealing without a common causes intermittent faults that look like equipment problems.

Verification

- 1 Power up and confirm 24 to 28 VAC between R and C at the thermostat subbase.
- 2 Set the thermostat configuration to match the equipment before the first call (system type, stages, O/B, fan with heat).
- 3 Jumper test from the subbase with the thermostat removed: R to G should run the indoor blower only.
- 4 R to Y, G and O should bring on cooling with the blower (B-type equipment: R to Y and G only).
- 5 R to Y and G should bring on compressor heat (B-type: add B). R to W2 / AUX should bring on the strips.
- 6 Reinstall the thermostat and run a full cycle in every mode the system supports.
- 7 Verify the changeover actually changes over - feel the discharge air, do not trust the display.
- 8 Record the configuration settings on the invoice and inside the furnace door.
- 9 Confirm the reversing valve: call for cool and feel for cold discharge air, then call for heat and confirm it turns warm. If they are swapped, O/B is wrong.
- 10 Force a defrost cycle if the board allows it and confirm auxiliary heat comes on to temper the supply air.

Common mistakes

Symptom	Check this first
Heats when you call cool, cools when you call heat	O/B reversed, either at the terminal or in the thermostat configuration.
Compressor never runs in heat, strips do everything	Electric heat landed on W instead of W2/AUX, or the stat is configured as conventional instead of heat pump.
Huge electric bill, strips always on	Aux configured to run with every heat call, or the balance point is set too high. Check the stat's aux staging settings.
Cooling fine, no heat at all	No W2/AUX landed and the reversing valve is stuck, or E is not jumpered where the air handler expects it.
Blank thermostat	No C wire. Heat pump stats draw more than a power-stealing stat can supply.
Service light on at the stat for no reason	L terminal picking up a fault from the outdoor board - read the outdoor board's flash code, do not chase the stat.